

Functional Requirements Specification

Food Delivery Platform – Production-Grade Specification

1. Refined Functional Requirements

FR-01: User Registration & Authentication

The system shall enable secure account creation and authentication.

- Registration shall require: Full Name, Email Address, Password (min 8 chars, complexity enforced), Verified Phone Number, Default Delivery Address.
- Email validation shall verify syntax and enforce a unique mail domain constraint (defined as the domain suffix following the @ symbol).
- Upon successful registration, the system shall generate a UUIDv4 System User ID. • Purpose of ID: Primary relational key across all system entities (orders, payments, profiles, audit logs). • Usage of ID: Routable identifier for transactional routing, session management, and crossservice dispatch.
- Authentication shall utilize email/password with cryptographically secure hashing (Argon2id/bcrypt), salted storage, and optional MFA integration.

FR-02: Browse Restaurants & Menus

The system shall provide a searchable, filterable catalog of restaurants and menu items.

- Filtering shall support: Food Category (e.g., Savory, Dessert), Cuisine Type (e.g., Pizza, Asian, Burger), and Distance-based radius.
- Delivery time estimation shall be computed per restaurant using: Geospatial distance, historical preparation averages, and real-time courier pool availability.
- The session architecture shall natively support multiple concurrent active orders per user account.

FR-03: Place Orders

The system shall enable item selection, cart management, and order finalization.

- Cart operations shall support quantity adjustment, item removal, and price recalculation with tax/delivery fee application.
- Checkout shall trigger order validation against restaurant availability and inventory state.
- Each finalized order instance shall be strictly bound to a single restaurant entity.

FR-04: Special Requests

The system shall capture and route customer-specific instructions.

- Two structured input fields shall be provided during checkout: (1) Culinary Modifications (ingredients, allergies, prep style), (2) Delivery Protocols (contact preference, access instructions, drop-off directives).
- Metadata shall be persisted with the order record and rendered prominently in restaurant prep tickets and courier delivery manifests.

FR-05: Payment Processing

The system shall facilitate secure, multi-modal payment collection.

- Supported Payment Methods (3): (1) Cash on Delivery (COD), (2) Credit/Debit Card via Gateway, (3) Digital Wallets/E-Wallets.
- Payment Channels: Mobile App, Web Client, In-App Secure Checkout.
- Processing Mechanisms: Tokenized transaction routing, PCI-DSS compliant gateway integration, asynchronous confirmation states, automated ledger reconciliation. • Online payments shall lock order validation until successful payment acknowledgment is received.

FR-06: Real-Time Order Tracking

The system shall provide an interactive, state-driven tracking interface.

- UI shall be map-based, WebSocket/SSE driven, updating at configurable intervals (5–10s).
- State Machine Coverage: Placed → Accepted → Preparation → Ready for Pickup → Picked Up → In Transit → Delivered.
- Interface shall display current state, agent location (post-pickup), progress indicators, and dynamic ETA recalculation.

FR-07: Notification System

The system shall dispatch multi-channel alerts for order milestones.

- Notifications: System-generated push, SMS, or in-app messages triggered by state transitions.
- Mobile Alerts: OS-level high-priority push notifications (e.g., "Order #1234 is out for delivery").
- Order Updates: Milestone progression alerts (e.g., "Restaurant accepted", "Meal preparation started").
- Delivery Tracking Alerts: Proximity/ETA alerts (e.g., "Courier 5 mins away", "Deliverer arrived at location").

FR-08: Rating & Review System

The system shall collect post-delivery feedback for restaurants and couriers.

- Two independent review entities shall be generated upon Delivered state completion: (1) Restaurant (Food Quality/Packaging), (2) Courier (Service/Punctuality).
- Each shall support 1–5 numerical scoring + optional text review.
- Data shall be timestamped, stored immutably, and aggregated for rolling average computation per entity.

FR-09: Order Management (Restaurants)

The system shall provide restaurants with order reception and preparation management.

- Dashboard shall display incoming orders with payload: Customer Address, Detailed Item List, Special Requests/Preferences, Assigned Courier Contact Info.
- Restaurant shall be required to input/confirm Estimated Preparation Time, which feeds dispatch scheduling algorithms.
- Internal queue prioritization (FCFS, priority, custom) shall be managed locally by restaurant operations; system remains agnostic to internal kitchen routing.

FR-10: Update Order Status (Restaurants)

The system shall enable restaurants to transition order states and communicate delays.

- Manual state transitions (e.g., Preparation → Ready for Pickup) shall trigger immediate mobile push to assigned courier device.

- Integrated communication channel (secure in-app messaging or masked VoIP) shall be available for delay reporting, stock adjustments, or route clarification.

FR-11: Delivery Assignment

The system shall automatically allocate delivery agents to ready orders.

- Primary Logic: Proximity to restaurant location (Euclidean/Network distance).
- Optimization Model: Weighted multi-objective function combining: Proximity (40%), Agent Load/Capacity (20%), Historical On-Time Rate (20%), Predicted Travel Time via Live Traffic (20%).
- Justification: Aligns with Operations Research Vehicle Routing Problem with Time Windows (VRPTW) principles, minimizing system-wide latency and maximizing throughput.
- Constraint Enforcement: One agent assigned per order instance.

FR-12: Order History & Analytics

The system shall store transactional records and generate user-level analytics.

- Persistent chronological log accessible via dedicated UI.
- Aggregated metrics: Order frequency, cuisine preference, average order value, delivery latency.
- Dataset shall serve as input for personalization engines and enable user re-ordering/auditing.

FR-13: Transaction, Refund & Complaints Processing

The system shall manage financial settlement and dispute resolution workflows.

- Secure payment routing from customer to merchant.
- Restaurant Payout Methods: Scheduled bank transfers, direct deposit to registered merchant wallets, platform balance payouts.
- Dispute Resolution Framework: Authorized admins may process partial/full refunds governed by: (1) Platform Service Agreements/Contracts, (2) Operational Refund Policies, (3) Applicable Consumer Protection Legal Frameworks.
- All financial adjustments shall generate immutable audit trails and trigger ledger reconciliation.

FR-14: Delivery Tracking & Updates (Agent)

The system shall enable couriers to report operational milestones and exceptions.

- Supported Status Updates: Picked Up, In Transit, Traffic Delay, ETA Adjusted, Delivered.
- Exception Logging: Couriers shall log events (e.g., "Road Closure", "Customer Unresponsive"). Logs propagate as real-time alerts to customer and restaurant.
- Distinct from automated GPS tracking; focuses on manual/operational state confirmation and exception reporting.

FR-15: Favorites, Recommendations & Real-Time Coordination (NEW)

The system shall implement preference management, algorithmic recommendations, and bidirectional availability synchronization.

- Favorites Module: Manual bookmarking of preferred restaurants.
 - Recommendation Engine: Surfaces restaurants based on: (1) Geospatial proximity, (2) Historical order behavior/cuisine affinity, (3) Projected delivery time scenarios.
 - Real-Time Communication: Inventory/availability sync to prevent stockouts. Order confirmation requires mutual acknowledgment (system push + restaurant accept). Postplacement modification/cancellation permitted within configurable window, triggering automatic restaurant halt notification and dispatch logic reset.
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2. Constraints (Strictly Enforced)

1. C-01: One order instance = exactly one restaurant.
 2. C-02: One delivery agent = one restaurant per active order instance.
 3. C-03: User accounts may maintain multiple simultaneous active orders across distinct restaurants.
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3. Scenario Formalization

S-01: Restaurant Connectivity Failure

- Detection: Order Sent state exceeds TTL (e.g., 90s) without restaurant acknowledgment.
- Fallback: Delivery agent initiates offline pickup protocol via cached order payload. System requires explicit Confirm/Reject UI toggle upon agent device reconnection or postreconciliation sync.
- Logging: Connectivity loss, TTL breach, and fallback activation recorded for SLA monitoring.

S-02: Delivery Assignment Failure

- Detection: Order remains in Ready for Pickup state beyond configurable threshold without agent assignment.
- Fallback: System escalates to Unassigned Priority Queue. System Administrator executes manual assignment via override dashboard. Audit trail captures assignment latency and manual intervention reason code.

4. Analytical Insights

Category	Details
Conflicts	FR-03/FR-02 allow multiple simultaneous orders, but C-01/C-02 enforce strict 1:1 mapping per order/agent. Resolution: Handled via session isolation and state machine partitioning.
Constraints	UUID primary keys, PCI-DSS routing, TTL-based fallbacks, and immutable audit logs introduce baseline compliance overhead.
Risks	1. GPS drift/telemetry loss may desync tracking UI. 2. Payment gateway timeouts during checkout. 3. Manual refund abuse. Mitigation: Idempotent APIs, webhook reconciliation, evidence-locked refund UI.
Observations	State machine design (Placed→Delivered) must be strictly enforced. Recommendation engine (FR-15) requires explicit user consent flags for privacy compliance.

5. Ambiguity Report

Per strict instruction: Ambiguous original phrasings are logged below.

Original Requirement	Ambiguity Explanation
Req-1: "gives each stakeholder...unique id"	"Stakeholder" scope undefined. "Some sort of unique id" lacks technical specificity.

Req-4: "send a notification via contact method or app"	Does not specify whether system should enforce or merely relay instruction. Contact method fallback undefined.
Req-9: "sorting out the protocol...fcfs...or priority queue"	Mixes functional requirements with internal restaurant operational policy.
Original Requirement	Ambiguity Explanation
Req-13: "Transaction, Refund Processing&complaints"	Punctuation/scope merging implies single process. No boundary between automated and manual handling.

6. Summary Report

Metric	Count / Example
Restrictions Preserved	3 (C-01, C-02, C-03)
Ambiguous Requirements Logged	4 (Req-1, Req-4, Req-9, Req-13)
Added Requirements	1 (FR-15: Favorites, Recommendations & Real-Time Coordination)
Future Features Documented	5 (ML offers, Manual fallback recs, Crowd estimation, Multi-order consolidation, Cloud automation)
Scenarios Formalized	2 (S-01: Restaurant Internet Failure, S-02: Delivery Assignment Failure)

Future Features Section (Documented for Phase 3+ Roadmap)

- ML-Based Offer Notifications: Predictive push campaigns using clustering algorithms on FR12 historical data.
- Manual Fallback Recommendation System: Rule-based heuristic engine if ML pipeline fails.
- Restaurant Crowd Estimation: IoT/POS integration to infer kitchen load, dynamically adjusting FR-02 delivery time estimates.
- Order Consolidation (Same Restaurant): Batch routing logic allowing single courier pickup for multiple user orders.
- Cloud Automation (FR-14): Serverless event-driven pipelines for status logging, exception routing, and telemetry aggregation.

Non-Functional Requirements Specification

Refined Non-Functional Requirements

NFR-01: Performance Efficiency

Description: The system shall meet defined latency, throughput, and resource utilization thresholds under concurrent load.

Specification:

- **Response Time (p95/p99):** Critical API endpoints (authentication, cart, payment initiation) $\leq 500\text{ms}$ at p95, $\leq 800\text{ms}$ at p99 under baseline load. UI render time for core views $\leq 2.0\text{s}$.
- **Throughput:** Minimum sustained capacity of 1,000 concurrent authenticated sessions; scalable to $\geq 5,000$ RPS during peak windows.
- **Resource Utilization:** CPU/Memory consumption $\leq 75\%$ average under sustained load; autoscale triggers at $\geq 80\%$ for 5-minute rolling average.
- **Validation:** Continuous load testing (k6/JMeter), APM telemetry (Datadog/New Relic), and performance regression gates in CI/CD.

NFR-02: Usability & Accessibility

Description: The system shall provide an intuitive, consistent, and accessible interface across supported platforms.

Specification:

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Learnability: First-time users shall complete core flow (browse → cart → checkout) in ≤ 3 primary interactions with ≤ 2 recoverable errors.

- **Accessibility:** WCAG 2.1 Level AA compliance: color contrast $\geq 4.5:1$, keyboard navigation parity, screen reader semantic markup, touch target $\geq 44 \times 44$ px.
- **Consistency:** Unified design system with standardized component library, interaction patterns, and error messaging taxonomy.
- **Validation:** Automated a11y scans (axe-core), heuristic evaluation against Nielsen's 10 usability heuristics, quarterly SUS (System Usability Scale) scoring ≥ 80 .

NFR-03: Security & Cryptography

Description: The system shall enforce cryptographic standards, secure data handling, and rolebased access controls.

Specification:

- **Encryption at Rest:** AES-256-GCM or ChaCha20-Poly1305 for all stored sensitive data (PII, tokens, credentials).
- **Encryption in Transit:** TLS 1.3+ with HSTS, OCSP stapling, and forward secrecy enforced on all external/internal endpoints.
- **Key Management:** HSM-backed or cloud KMS integration with automated 90-day rotation, separation of duties for key escrow.
- **Access Control:** RBAC/ABAC model enforcing least privilege. MFA mandatory for administrative/merchant dashboards. JWT/OAuth 2.1 with short-lived access tokens + refresh token rotation.

NFR-04: Data Privacy & Compliance

Description: The system shall comply with regional data protection regulations and enforce user data lifecycle controls.

Specification:

- **Regulatory Alignment:** GDPR (EU), CCPA/CPRA (California), and equivalent local frameworks. Data processing mapped to lawful basis documentation.
- **Data Subject Rights:** Automated DSAR pipeline for data access/export within 30 days. Rightto-erasure execution within 30 days of validated request, with immutable audit logging of deletion.

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- **Data Minimization:** Collection limited to FR-01 required fields. Pseudonymization applied to analytics/training datasets.

Consent Management: Explicit opt-in/opt-out tracking for marketing/tracking cookies, persisted and versioned.

NFR-05: Availability & Uptime SLA

Description: The system shall maintain continuous operational availability with defined fault boundaries. **Specification:**

- **SLA Target:** 99.95% monthly uptime (≤ 21.6 minutes allowable downtime/month). Excludes scheduled maintenance windows.
- **Architecture:** Multi-AZ/Multi-Region deployment with active-passive or active-active failover. Health checks with automated pod/instance replacement.
- **Monitoring:** Synthetic transaction monitoring, real-time error budget tracking (SRE methodology), PagerDuty/Alertmanager escalation policies.

NFR-06: Maintainability & Backup/Recovery

Description: The system shall support continuous updates, fault-tolerant operations, and verified data recovery.

Specification:

- **Deployment Strategy:** Zero-downtime releases via blue-green or canary deployments. Automated rollback on health degradation $>$ threshold.
- **Patch Management:** Critical CVEs remediated within 72 hours of disclosure; non-critical within 30 days. Dependency scanning in CI pipeline.
- **Backup & Recovery:** Daily incremental + weekly full encrypted backups. $RPO \leq 15$ minutes, $RTO \leq 60$ minutes. Quarterly disaster recovery drills with documented success metrics.
- **Fault Tolerance:** Graceful degradation of non-critical services (e.g., reviews, recommendations) without core checkout/tracking failure.

NFR-07: Scalability & Elasticity

Description: The system shall dynamically scale to accommodate growth in users, orders, and data volume without architectural redesign.

Specification:

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- **Horizontal Scaling:** Stateless service tier auto-scales based on CPU/memory thresholds and queue depth metrics.
Data Tier Scaling: Read replicas for analytical queries; connection pooling (PgBouncer/ProxySQL); horizontal partitioning/sharding for order history beyond baseline capacity.
- **Load Leveling:** Asynchronous event streaming (Kafka/RabbitMQ) for order placement, notifications, and status updates to decouple producers/consumers.
- **Capacity Planning:** Linear scaling up to 10x baseline load with $\leq 10\%$ latency degradation; documented runbooks for capacity expansion.

NFR-08: Reliability & Fault Tolerance

Description: The system shall maintain operational continuity during third-party or infrastructure failures through defined degradation patterns.

Specification:

- **Circuit Breaker Pattern:** Implemented on all external dependencies (GPS, payment gateways, SMS providers). Open state after configurable failure threshold (e.g., 5 errors/10s), fallback to cached/last-known state.
 - **Retry & Idempotency:** Exponential backoff with jitter for transient failures. Idempotent request keys for payment, order creation, and status updates.
 - **Client Resilience:** Unhandled exceptions routed to safe UI states. Memory leak detection and crash analytics integrated (Sentry/Crashlytics).
 - **Observability:** OpenTelemetry instrumentation for distributed tracing, structured JSON logging, metric correlation. Error budgets tracked per service SLO.
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Constraints & Architectural Patterns

Constraint	Specification
Statelessness	All application tiers must be stateless; session state externalized to Redis/DynamoDB.
Idempotency	All mutating APIs require idempotency keys to prevent duplicate orders/payments on retry.
Decoupling	Event-driven architecture enforced for FR-06/FR-10/FR-14 status propagation.
Zero-Trust Networking	mTLS for inter-service communication; WAF + API gateway for external ingress.

Analytical Insights

Category	Details
Conflicts	NFR-01 (performance) vs. NFR-03/04 (security/compliance): Encryption overhead and audit logging can increase latency. <i>Resolution:</i> Hardwareaccelerated crypto (AES-NI), async audit log batching, and connection pooling mitigate impact.
Constraints	$RPO \leq 15min + RTO \leq 60min$ requires synchronous cross-AZ replication and warm standby infrastructure. Increases baseline cloud cost; must be scoped to production tier only.
Risks	1. GPS fallback (NFR-08) may mislead users if last-known location is stale > 5min. 2. Auto-scaling thrashing during traffic spikes. 3. DSAR pipeline failure causing compliance violations. <i>Mitigation:</i> TTL caching on location data, predictive scaling + cooldown windows, automated DSAR queue with manual escalation.
Observations	NFRs are interdependent: Scalability (NFR-07) enables Performance (NFR-01); Fault Tolerance (NFR-08) enables Availability (NFR-05). Implementation must follow SLO-driven prioritization, not feature-driven.

Ambiguity Report

Original phrasings mapped to vagueness sources per ISO/IEC 25010 measurement guidelines.

Original Requirement	Ambiguity Explanation
Performance: "respond within a short time"	"Short time" is non-quantifiable. Lacks percentile targets (p50/p95/p99), scope (API vs UI), and load context.
Usability: "simple and intuitive UI"	Subjective terms without measurable UX metrics, accessibility baseline, or user journey success criteria.
Availability: "minimal downtime"	"Minimal" undefined. No SLA percentage, exclusion windows, or failover architecture specified.

Maintenance: "up to date always"	"Always" is physically impossible. No patch SLA, vulnerability response window, or deployment strategy defined.
Original Requirement	Ambiguity Explanation
Scalability: "support increasing numbers"	Lacks scaling boundaries, architectural patterns (horizontal/vertical), data tier strategy, or performance degradation thresholds.

Summary Report

Metric	Count / Example
Refined NFRs	8 (NFR-01 to NFR-08, logically split per ISO 25010 quality model)
Quantifiable Metrics Added	14 (p95/p99 latency, RPS, SLA %, RPO/RTO, WCAG level, SUS score, rotation cycles, etc.)
Industry Standards Applied	ISO/IEC 25010, NIST SP 800-57/81, PCI-DSS v4.0, GDPR/CCPA, WCAG 2.1, OWASP ASVS, SRE/SLO Framework
Ambiguous Original Phrases Logged	5 (Performance time, Usability simplicity, Downtime minimal, Updates always, Scalability undefined)
Architectural Constraints	4 (Statelessness, Idempotency, Event Decoupling, Zero-Trust)